ARASH BABAMIRI

PhD Candidate - Building Energy Modeling

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WORK EXPERIENCE

PhD Student - Building Energy Modeling

Sep 2023 - Present   Dublin City University, Dublin, Ireland

- Conducting PhD research in building energy modeling as part of the SEAI-funded ENABLE project (retrofit of residential buildings in Ireland), supervised by Dr. Reihaneh Aghamolaei.
- Designed a BIM-based adaptive sparse Polynomial Chaos Expansion (PCE) surrogate framework for efficient uncertainty quantification in building energy simulation, applied to four Irish residential archetypes (approximately 500 EnergyPlus runs per archetype vs. an approximately 16,000-run Monte Carlo benchmark, coefficient of determination ≥ 0.99 for EUI, approximately 8× cost reduction).
- Published results in the Journal of Energy and Buildings (Impact Factor 8.0); developed methods for modelling building energy consumption before and after retrofitting.
- Presented research at international conferences, including EC3 2026 (Greece) and the IBPSA-Ireland National Student Symposium.

Module Demonstrator / Teaching Assistant

Sep 2024 - Present   Dublin City University, Dublin, Ireland

- Facilitated lab sessions, tutorials, and practical coursework demonstrations for undergraduate engineering modules: Design and CAD (2024-2026), Numerical Problem (2025-2026), Thermodynamics: Energy (2024-2025), and Engineering Mechanics: Dynamics (2024-2025).

Researcher - Biomechanics (Freelance)

Jan 2022 - Dec 2022   Remote

- Collaborated with a research team on biomechanics projects, applying fluid mechanics and CFD analysis (Ansys Fluent).
- Co-authored three papers published in international conferences and journals on particle deposition in tracheobronchial airway models and dry powder inhaler aerosol deposition.

Mechanical Designer / Industrial Drafter

2021 - 2023   D.M.O. - Diesel Movaled Omid, Kashan, Iran

- Designed chassis and canopy covers for diesel generators; created detailed 3D models and technical drawings and managed assembly processes.
- Executed sheet metal fabrication and laser cutting projects; learned ProNest 2021 nesting software and the SolidWorks Sheet Metal module.
- Collaborated with cross-functional teams to streamline manufacturing workflows and automated the design process, reducing manual effort by up to 80%.

PROFILE

Mechanical engineer and PhD candidate specialising in building energy modeling, with a background spanning thermodynamic analysis of heat recovery and renewable energy systems, BIM, and machine learning. Combines hands-on CAD/design experience (SolidWorks, AutoCAD, Revit MEP) with coding and data-analysis skills (Python, surrogate modelling, uncertainty quantification). Fast learner and diligent researcher with a strong record of peer-reviewed publications and conference presentations.

SKILLS

- **Building Performance & BIM:** EnergyPlus, Revit MEP, Dynamo (Python scripting), HVAC Design, weather data analysis, IES Virtual Environment
- **Design & CAD:** SOLIDWORKS, AutoCAD, Design-Build, sheet metal design, technical drawing
- **Programming & Data:** Python, Machine Learning, Generative AI, Data Analysis, Statistics
- **Research & Documentation:** Technical documentation, technical reviews, continuous improvement
- **Professional & Academic:** University teaching/demonstration, teamwork, time management, Microsoft PowerPoint, professional communication

RESEARCH INTERESTS

- Building energy modeling
- Energy efficiency and retrofit
- Uncertainty quantification
- Surrogate modelling
- BIM and computational workflows
- Machine learning
- Renewable energy systems

WORK EXPERIENCE

Researcher

Jan 2021 - Sep 2023 University of Kurdistan, Sanandaj, Iran

- Collaborated with Dr. Ebrahimi, Professor of Heat Recovery and CCHP Systems, as a research assistant, writing scientific articles on the energy and thermodynamics of integrated renewable systems.
- Co-authored work on hybrid solar collector-chimney, thermoelectric and wind-turbine systems, and a micro-trigeneration cycle based on a Stirling engine, organic Rankine cycle, and adsorption chiller.

Industrial Designer

2020 - 2021 NTGC, Iran

- Performed reverse engineering of industrial machine components for a glass-manufacturing company with 2,000+ employees.
- Developed 3D models and industrial drawings using SolidWorks.

Mechanical Design Engineer & Industrial Drafter

Feb 2019 - Nov 2021 Noritazeh, Tehran Province, Iran

- Designed new equipment and reverse-engineered broken components as part of the design team; followed up on parts manufacturing.
- Deepened expertise in mechanical design and SolidWorks; translated design concepts into manufacturing- and construction-ready documentation.

Mechanical Designer / HVAC Designer Engineer

2018 - 2019 Pendar, Iran

- Designed and sized HVAC systems (ductwork, piping, equipment) for industrial buildings using AutoCAD and Revit MEP.
- Coordinated with multidisciplinary teams to ensure MEP integration and compliance with ASHRAE 90.1 and local codes.
- Prepared technical documentation including equipment specifications, BOQs, and vendor submittal reviews.

QC Manager

Jul 2017 - Apr 2018 Azmapayam Sarmad, Tehran Province, Iran

- Managed quality control processes, documentation, and technical reviews.

EDUCATION

PhD, Building Energy Modelling

Sep 2023 - Present Dublin City University

- Designing and developing methods for modelling building energy consumption before and after retrofitting; publishing results in high-ranking scientific journals.

Master’s Degree, Mechanical Engineering

Malek Ashtar University of Technology

- Ranked 4th of 15 students.
- Thesis: Stress and fatigue analysis of car chassis.

BSc, Mechanical Engineering

2011 - 2015 University of Kurdistan

- Ranked 2nd among 52 students.
- Thesis: Energy and exergy analysis of a hybrid air conditioning system.

LEADERSHIP & PROFESSIONAL ACTIVITIES

- **Organizing Committee - IBPSA-Ireland National Student Symposium 2026, DCU:** Co-organized a national symposium for the Irish building performance simulation community, including student presentations, poster sessions, and a Thesis-in-3-Minutes (3MT) competition (3-4 June 2026).
- **Presenter - IBPSA Ireland seminar series:** Presented PhD research and engaged with the Irish building performance simulation community.
- **Speaker - EC3 2026, Greece:** Presented PhD research as part of the SEAI-funded ENABLE project.

PUBLICATIONS

Google Scholar

Journal Articles

- **Babamiri Naamrudi, A.**; Aghamolaei, R. "A BIM-based adaptive surrogate framework for efficient uncertainty quantification in building energy simulation." *Energy and Buildings*, 370, 118098, 2026.
- Shakeri, A.; Asadbagi, P.; **Babamiri Naamrudi, A.** "Techno-economic, techno-environmental assessments, and deep learning optimization of an integrated system for CO_2 capturing from a gas turbine: Tehran case study." *Energy*, 306, 132438, 2024.
- Momeni Larimi, M.; **Babamiri, A.**; Biglarian, M.; Ramiar, A.; Tabe, R.; Inthavong, K.; Farnoud, A. "Numerical and Experimental Analysis of Drug Inhalation in Realistic Human Upper Airway Model." *Pharmaceuticals*, 16(3), 406, 2023.
- **Babamiri, A.**; Ahookhosh, K.; Abdollahi, H.; Taheri, M. H.; Cui, X.; Nabaei, M.; Farnoud, A. "Effect of laryngeal jet on dry powder inhaler aerosol deposition: a numerical simulation." *Computer Methods in Biomechanics and Biomedical Engineering*, 26(15), 1859–1874, 2023.
- Kazemi, K.; Ebrahimi, M.; Lahonian, M.; **Babamiri, A.** "Micro-scale heat and electricity generation by a hybrid solar collector-chimney, thermoelectric, and wind turbine." *Sustainable Energy Technologies and Assessments*, 53, 102394, 2022.
- **Babamiri, A.**; Gharib, M.; Ebrahimi, M. "Multi-criteria evaluation of a novel micro-trigeneration cycle based on α-type Stirling engine, organic Rankine cycle, and adsorption chiller." *Energy Conversion and Management*, 253, 115162, 2022.
- Wheatley, G.; **Babamiri, A.**; Philippa, B. "Vibration analysis of an Airlie beach house: a case study in Australia." *Scientific Journal of Silesian University of Technology. Series Transport*, 114, 179–192, 2022.
- Mahdian, A.; **Babamiri, A.**; Shahriari, B. "Weld Distortion and Residual Stresses in Aluminum Hollow Section T-Joint." *International Journal of Advanced Design and Manufacturing Technology*, 15(1), 109–114, 2022.
- **Babamiri, A.**; Hadian, S.; Wheatley, G. "Fatigue and Stress Analysis of a Load Carrying Car Chassis with Reinforced Joint Using Finite Element Method." *Revista GEINTEC: Gestão, Inovação e Tecnologias*, 11(4), 2156–2176, 2021.

Conference Papers

- Abdollahi, H.; **Babamiri, A.**; Ahookhosh, K.; Farnoud, A.; Nabaei, M. "Effects of Inhalation Flow Rate on Particle Deposition and Flow Structure in a Model of Tracheobronchial Airway." *2021 28th National and 6th International Iranian Conference on Biomedical Engineering (ICBME)*, IEEE, pp. 101–106, 2021.
- **Babamiri, A.**; Lahonian, M.; Ebrahimi, M. "Design and exergy analysis of compressive-evaporative hybrid cooler" (in Persian). *4th National and 2nd International Conference on Applied Research in Electrical, Mechanical and Mechatronics Engineering*, Tehran, Iran, 2017.
- Ebrahimi, M.; Lahonian, M.; Ahookhosh, K.; **Babamiri, A.** "Design and optimization of a heat recovery system for a CHP system." *6th International Conference on Heating, Ventilating and Air Conditioning (ICHVAC-6)*, Tehran, Iran, 2015.

CERTIFICATIONS & TRAINING

- **Green Building Concepts Foundations** - LinkedIn Learning, Feb 2026
- **Nano Tips for Building Soft Skills with Elayne Fluker** - LinkedIn Learning, Jun 2025
- **Communicating with Confidence** - LinkedIn Learning, Jun 2025
- **Prompt Engineering: How to Talk to the AIs** - LinkedIn Learning, Jun 2025
- **Practice Exam for LEED Green Associate** - LinkedIn Learning, Mar 2025
- **Revit 2023: Essential Training for MEP** - LinkedIn Learning, Mar 2025
- **Dynamo for Revit Python Scripting** - LinkedIn Learning, Feb 2025
- **IES Virtual Environment Summer School** - IES Ltd.: Building Geometry Creation, Energy Analysis & Load Calculations, HVAC System Modelling, Daylight & Sunlight Modelling, Thermal Bridging & Parametric Simulations
- **English Proficiency Certificate** - Duolingo English Test, May 2023
- **GRE** - ETS, Sep 2021
- **Applied Machine Learning in Python** - Coursera, Apr 2021
- **Programming for Everybody (Getting Started with Python)** - Coursera, Mar 2021
- **TOEFL** - ETS, Jul 2020